

Practice Exercise 2

1. Variable Reassignment

Question:

Write a program that stores a person's name in a variable and prints it. Then, update the variable with a new name entered by the user and print the updated name.

2. Data Types

Question:

Write a program that declares and initialises variables of the following data types:

- A string
- An integer
- A float
- A boolean

Print the value and data type of each variable.

3. User Input

Question:

Write a program that prompts the user to enter a number. Convert the input to an integer and a float, and print both values along with their data types.

4. Calculations

Question:

Write a program that prompts the user to enter two numbers. Perform the following calculations and display the results:

- Addition
- Subtraction
- Multiplication
- Division
- Integer division
- Exponentiation
- Modulus (remainder)

5. Mixed-Type Expressions and Data Type Conversion

Question:

Write a program that demonstrates mixed-type arithmetic operations.

- Assign an integer and a float to two variables.
- Perform addition, subtraction, and multiplication on them.
- Convert the float to an integer and display the result.

6. Long Statements

Question:

Write a program that calculates the total cost of five different items.

- Store each item's price in a separate variable.
- Use line continuation (\) or parentheses to break the calculation into multiple lines.
- Print the total cost.

7. String Concatenation

Question:

Write a program that asks the user for their first name and last name.

- Concatenate the names with a space in between and print the full name.
- Print a message using concatenation to welcome the user.

8. Print Function

Question:

Write a program that prints the following details in a single print statement:

- Your name
- Your age
- Your favourite hobby

Separate each value using a comma without using multiple print statements.

9. Formatted Output with F-Strings

Question:

Write a program that prompts the user to enter:

- A product price
- A discount percentage

Use an f-string to display the original price, the discount amount, and the final price, formatted to two decimal places.

10. Magic Numbers and Named Constants

Question:

Write a program that calculates the final amount after interest on a bank balance.

- Define a constant for the interest rate (e.g., `INTEREST_RATE = 0.05`).
- Prompt the user to enter their balance.
- Calculate the interest earned and display the final balance.

Avoid using magic numbers.